

Solution to Ramanujan Challenge Problem 2.5: Efficient Rational Approximation of Catalan's Constant G

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August 2026

Abstract

We prove, for every one of the three terminal columns, that the matrix quotient in Problem 2.5 tends to Catalan's constant. The paper contains two exact proofs. The first relates a Meijer- G trajectory to the printed matrix by an explicit inverse-transpose identity. The second is independent of that special-function connection: an elementary positive triple integral produces a fast adjoint solution, while a rational companion and a contracting Miller cone select its projective limit. We also give the exact inverse Delannoy transform and the requested finite integral formula for its coefficient error. No finite-prefix guess or decimal recognition is used.

1 The matrix and the assertion

Let $M(n) = (m_{ij}(n))_{1 \leq i, j \leq 3}$ be the matrix in the question. For reference, its entries are

$$\begin{aligned} m_{11} &= (-2n - 5)(n + 3)^2(136n^4 + 1424n^3 + 5548n^2 + 9551n + 6141), \\ m_{12} &= 384n^6 + 6384n^5 + 44168n^4 + 162698n^3 + 336377n^2 + 369933n + 169011, \\ m_{13} &= -480n^4 - 4980n^3 - 19210n^2 - 32690n - 20730, \\ m_{21} &= (n + 2)^2(n + 3)^2(4n + 10)(48n^3 + 386n^2 + 1017n + 879), \\ m_{22} &= (n + 2)^2(-272n^5 - 3848n^4 - 21732n^3 - 61184n^2 - 85761n - 47808), \\ m_{23} &= (n + 2)^2(320n^3 + 2540n^2 + 6610n + 5640), \\ m_{31} &= (-4n - 10)(n + 2)^2(n + 3)^2(32n^4 + 302n^3 + 1037n^2 + 1530n + 813), \\ m_{32} &= (n + 2)^2(192n^6 + 2984n^5 + 19116n^4 + 64452n^3 + 120256n^2 + 117279n + 46476), \\ m_{33} &= (n + 2)^2(-16n^5 - 408n^4 - 2912n^3 - 8884n^2 - 12254n - 6240). \end{aligned}$$

Put

$$\mathcal{M}_N = M(0)M(1) \cdots M(N - 1), \quad A = \begin{pmatrix} 30921 & -32972 & 8240 \\ 33750 & -36000 & 9000 \end{pmatrix},$$

where the empty product is the identity, and define

$$A\mathcal{M}_N = \begin{pmatrix} P_{N,1} & P_{N,2} & P_{N,3} \\ Q_{N,1} & Q_{N,2} & Q_{N,3} \end{pmatrix}.$$

Finally,

$$G = \sum_{r=0}^{\infty} \frac{(-1)^r}{(2r + 1)^2}.$$

Theorem 1 (Main assertion). *For $j = 1, 2, 3$,*

$$\lim_{N \rightarrow \infty} \frac{P_{N,j}}{Q_{N,j}} = G.$$

The proof is completed in Theorem 19.

Lemma 2 (Catalan moment). *The improper integral is absolutely convergent and*

$$G = \int_0^1 \frac{-\log t}{1+t^2} dt, \quad \int_0^1 (-\log t) dt = 1.$$

Proof. The second identity follows by integration by parts. For $0 \leq t < 1$,

$$\frac{1}{1+t^2} = \sum_{r \geq 0} (-1)^r t^{2r}.$$

The partial sums have absolute value at most 2, while $-\log t$ is integrable on $(0, 1)$. Dominated convergence and $\int_0^1 t^{2r} (-\log t) dt = (2r+1)^{-2}$ give the first identity. $\square$

The following exact moment identity will also be used in the Delannoy appendix. It is not used to infer decay merely from a finite alternating sum.

Lemma 3 (Exact moment lift). *Let $\mathbf{p}_0$ and $\mathbf{q}_0$ be the two rows of A , set*

$$\mathbf{R}_0(X) = \mathbf{q}_0 - (1+X)\mathbf{p}_0, \quad R_{N,j}(X) = (\mathbf{R}_0(X)\mathcal{M}_N)_j.$$

Then $R_{N,j}(X) = Q_{N,j} - (1+X)P_{N,j}$ and

$$Q_{N,j} = \int_0^1 (10-18x)R_{N,j}(x) dx, \tag{1}$$

$$P_{N,j} = \int_0^1 (6-12x)R_{N,j}(x) dx, \tag{2}$$

$$GQ_{N,j} - P_{N,j} = \int_0^1 \frac{-\log t}{1+t^2} R_{N,j}(t^2) dt. \tag{3}$$

Proof. The displayed affine formula for $R_{N,j}$ follows directly from the definition. If $R(X) = a+bX$, then the four elementary moments

$$\int_0^1 (10-18x) dx = 1, \quad \int_0^1 x(10-18x) dx = -1, \quad \int_0^1 (6-12x) dx = 0, \quad \int_0^1 x(6-12x) dx = -1$$

give (1) and (2). Lemma 2 gives

$$\int_0^1 \frac{-\log t}{1+t^2} (Q - (1+t^2)P) dt = GQ - P,$$

which is (3). The logarithmic endpoint is integrable, so these are ordinary improper Lebesgue integrals. $\square$

2 Inverse-transpose selection

We record the standard form of the recessive-solution principle needed below. It is a consequence of the Poincaré–Perron theorem in its higher-order (discrete Levinson) form; see [1].

Proposition 4 (Recessive recognition and Miller selection). *Suppose a row companion system $X_{n+1} = X_n C(n)$ satisfies $C(n) = C_\infty + O(n^{-1})$, where C_∞ is invertible and diagonalizable. Suppose C_∞ has a simple eigenvalue μ such that*

$$0 < |\mu| < \min\{|\lambda| : \lambda \in \text{spec}(C_\infty), \lambda \neq \mu\}.$$

If a nonzero scalar solution F_n has $\limsup |F_n|^{1/n} \leq |\mu|$, it spans the unique recessive solution line. Writing

$$\Phi_C(N) = C(0) \cdots C(N-1), \quad \mathbf{F}_0 = (F_0, \dots, F_{r-1})^T,$$

one has

$$[\Phi_C(N)^{-T} w_N] \longrightarrow [\mathbf{F}_0]$$

for every terminal vector w_N whose coefficient in the corresponding dominant dual direction is nonzero.

Moreover, if $T(n)U(n+1) = U(n)C(n)$ with rational invertible $U(n)$, then

$$[\Phi_T(N)^{-T} e_j] \longrightarrow [U(0)^{-T} \mathbf{F}_0] \tag{4}$$

whenever $U(N)^T e_j$ has nonzero recessive-dual connection.

Proof. The spectral gap splits the asymptotic solution space into the one-dimensional μ -space and its complementary block. Discrete Levinson asymptotics show that every solution outside the first space has strictly larger exponential rate. This proves the recognition statement. Inverting and transposing a fundamental product replaces each exponential factor λ^N by λ^{-N} ; hence the recessive primal line becomes the unique dominant dual line. This proves the projective assertion.

Finally, multiplication of the coboundary identities gives

$$\Phi_T(N) = U(0)\Phi_C(N)U(N)^{-1}, \quad \Phi_T(N)^{-T} = U(0)^{-T}\Phi_C(N)^{-T}U(N)^T.$$

Rational gauges have at most polynomial growth along the positive integers and therefore do not alter an exponentially isolated line. This gives (4). $\square$

3 An exact Meijer–G trajectory

We use the Barnes convention of [2, Sec. 16.17],

$$G_{p,q}^{m,r} \left(z \left| \begin{matrix} a_1, \dots, a_p \\ b_1, \dots, b_q \end{matrix} \right. \right) = \frac{1}{2\pi i} \int_L \frac{\prod_{j=1}^m \Gamma(b_j - s) \prod_{j=1}^r \Gamma(1 - a_j + s)}{\prod_{j=m+1}^q \Gamma(1 - b_j + s) \prod_{j=r+1}^p \Gamma(a_j - s)} z^s ds.$$

For $n \geq 0$ define

$$F_n(z) = G_{4,4}^{4,2} \left(z \left| \begin{matrix} -1-n, -1-n, n+\frac{5}{2}, n+3 \\ \frac{3}{2}, \frac{1}{2}, 1, 0 \end{matrix} \right. \right), \quad F_n = F_n(1), \tag{5}$$

and set $\Theta = z d/dz$ and

$$Y_n = (F_n, (\Theta F_n)(1), (\Theta^2 F_n)(1)).$$

Lemma 5 (Differential reduction). *At $z = 1$,*

$$(\Theta^3 F_n)(1) = -\frac{(n+2)^3(2n+3)}{7}F_n + \frac{2n^2+8n+5}{14}(\Theta F_n)(1) + \frac{8n^2+30n+39}{14}(\Theta^2 F_n)(1). \quad (6)$$

Proof. The Meijer- G differential equation for (5) is

$$[z(\Theta + n + 2)^2(\Theta - n - \frac{3}{2})(\Theta - n - 2) - (\Theta - \frac{3}{2})(\Theta - \frac{1}{2})(\Theta - 1)\Theta] F_n(z) = 0.$$

At $z = 1$ the fourth-order terms cancel. Expansion and division by $7/2$ gives (6). $\square$

Define $T(n) = (t_{ij}(n))$ by

$$\begin{pmatrix} \frac{4(n+2)(17n^3+111n^2+240n+171)}{(n+1)(n+3)(2n+3)(2n+5)} & \frac{(n+2)(24n^2+101n+102)}{(n+1)(2n+3)} & \frac{(n+2)(2n+5)(16n^2+81n+90)}{2(n+1)(2n+3)} \\ \frac{96n^4+780n^3+2384n^2+3273n+1723}{(n+1)(n+2)(n+3)(2n+3)(2n+5)} & \frac{68n^3+398n^2+778n+523}{2(n+1)(n+2)(2n+3)} & \frac{96n^4+884n^3+2970n^2+4360n+2403}{4(n+1)(n+2)(2n+3)} \\ \frac{-5(24n^2+117n+143)}{(n+1)(n+2)(n+3)(2n+3)(2n+5)} & \frac{-5(16n+41)}{2(n+1)(n+2)(2n+3)} & \frac{8n^3-44n^2-478n-801}{4(n+1)(n+2)(2n+3)} \end{pmatrix}.$$

Lemma 6 (Exact contiguous trajectory). *For every integer $n \geq 0$,*

$$Y_{n+1} = Y_n T(n).$$

Proof. In the Barnes integrand for F_n , the n -dependent factor is

$$\frac{\Gamma(n+2+s)^2}{\Gamma(n+\frac{5}{2}-s)\Gamma(n+3-s)}.$$

Its quotient after $n \mapsto n+1$ is

$$\frac{(n+2+s)^2}{(n+\frac{5}{2}-s)(n+3-s)}.$$

Apply this identity also after multiplication by s and s^2 . Gamma recurrence and the cubic reduction (6) reduce the three resulting Barnes integrals to the basis $1, s, s^2$; the three coefficient columns are exactly the displayed columns of $T(n)$. All operations are identities in $\mathbb{Q}(n, s)$ after the gamma factors are cancelled. They are first valid on a Barnes contour for $0 < z < 1$; gamma decay permits contour shifting and differentiation. Dominated passage to $z = 1$ gives the assertion. Thus this is an exact contiguous calculation, not a check of finitely many values. $\square$

Put

$$\sigma(n) = -2(n+2)^2(n+3)^2(2n+5)(2n+7)^2. \quad (7)$$

Lemma 7 (Inverse-transpose certificate). *For the printed matrix $M(n)$ and the displayed $T(n)$,*

$$T(n)^T M(n) = \sigma(n) I_3. \quad (8)$$

Consequently, if $\Phi_T(N) = T(0) \cdots T(N-1)$ and $H_N = \prod_{m=0}^{N-1} \sigma(m)$, then

$$\mathcal{M}_N = H_N \Phi_T(N)^{-T}, \quad H_N = (-16)^N (2)_N^2 (3)_N^2 (\frac{5}{2})_N (\frac{7}{2})_N^2. \quad (9)$$

Proof. Substitution of the nine displayed rational functions and the nine polynomials defining $M(n)$ gives (8) after clearing the nonzero denominator $(n+1)(n+2)(n+3)(2n+3)(2n+5)$. This is a nine-entry polynomial identity in $\mathbb{Q}[n]$. Multiplying it for $0 \leq n < N$ gives the first identity in (9); factoring each linear term in (7) gives the stated Pochhammer product. $\square$

4 Companion form and exact connection data

Let $e_1 = (1, 0, 0)^T$ and define the Krylov matrix

$$U(n) = [e_1, T(n)e_1, T(n)T(n+1)e_1]. \quad (10)$$

Lemma 8 (Krylov determinant). *Let*

$$D(n) = 3072n^7 + 61824n^6 + 531184n^5 + 2525522n^4 + 7175793n^3 + 12183785n^2 + 11446123n + 4589916.$$

Then

$$\det U(n) = -\frac{15D(n)}{2(n+1)(n+2)^3(n+3)^2(n+4)(2n+3)^2(2n+5)^2(2n+7)}.$$

In particular, $U(n)$ is invertible for every integer $n \geq 0$.

Proof. Expand the determinant using (10) and factor. Every coefficient of D is positive and all denominator factors are positive for $n \geq 0$. $\square$

Lemma 9 (Companion coboundary). *The matrix*

$$C(n) = U(n)^{-1}T(n)U(n+1)$$

is a row companion matrix, $T(n)U(n+1) = U(n)C(n)$, and

$$C(n) = C_\infty + O(n^{-1}), \quad C_\infty = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & -35 \\ 0 & 1 & 35 \end{pmatrix}.$$

Moreover

$$Y_n U(n) = (F_n, F_{n+1}, F_{n+2}). \quad (11)$$

Proof. The first two columns of the coboundary identity follow from the definition of $U(n)$; the last column defines the scalar recurrence, so $C(n)$ is companion. Exact rational simplification gives the displayed limit. Equation (11) follows from Lemma 6. $\square$

Lemma 10 (Spectral gap). *The characteristic polynomial of C_∞ is*

$$(\lambda - 1)(\lambda^2 - 34\lambda + 1).$$

Its roots are

$$\mu = 17 - 12\sqrt{2} = (\sqrt{2} - 1)^4, \quad 1, \quad \mu^{-1} = 17 + 12\sqrt{2}.$$

Thus μ is simple and has strictly smallest modulus. A right eigenvector for μ is

$$r_\mu = (17 + 12\sqrt{2}, -18 - 12\sqrt{2}, 1)^T. \quad (12)$$

Proof. This is direct factorization and multiplication by C_∞ . $\square$

Lemma 11 (All three terminal connections are nonzero). *For each $j = 1, 2, 3$, the vector $U(N)^T e_j$ has nonzero connection with the recessive dual mode in Proposition 4.*

Proof. Exact leading terms of the displayed rational matrix $U(n)$ give

$$\lim_{n \rightarrow \infty} \text{diag}(1, n, n^3)U(n) = L := \begin{pmatrix} 1 & 17 & 577 \\ 0 & 24 & 816 \\ 0 & -30 & -1020 \end{pmatrix}. \quad (13)$$

The three row pairings of L with (12) are

$$96(3 - 2\sqrt{2}), \quad 96(4 - 3\sqrt{2}), \quad 120(-4 + 3\sqrt{2}).$$

None is zero: otherwise $\sqrt{2}$ would be rational. Polynomial row scalings do not change whether the corresponding asymptotic coefficient vanishes. Hence all three terminal vectors satisfy the noncancellation hypothesis. $\square$

Let $\mathbf{F} = (F_0, F_1, F_2)^T$.

Lemma 12 (Barnes initial values). *The initial vector is*

$$\mathbf{F} = \sqrt{\pi} \begin{pmatrix} 150G - 128 \log 2 - \frac{146}{3} \\ \frac{24745}{32}G - \frac{14624}{5} \log 2 - \frac{823511}{33600} \\ \frac{7225281}{32}G - \frac{886784}{5} \log 2 - \frac{2818419551}{33600} \end{pmatrix}. \quad (14)$$

Proof. Insert $n = 0, 1, 2$ in the Barnes integrand in (5). Cancelling the two gamma quotients gives

$$\frac{\Gamma(\frac{1}{2} - s)\Gamma(-s)\Gamma(n + 2 + s)^2}{(\frac{3}{2} - s)_{n+1}(1 - s)_{n+2}}.$$

Residue expansion (first at $z = r$ with $0 < r < 1$ and then with $r \uparrow 1$) reduces each of the three cases by partial fractions to

$$\sum_{k \geq 0} \frac{(-1)^k}{2k+1} = \frac{\pi}{4}, \quad \sum_{k \geq 1} \frac{(-1)^{k-1}}{k} = \log 2, \quad \sum_{k \geq 0} \frac{(-1)^k}{(2k+1)^2} = G,$$

plus finite rational tails. Collecting the coefficients gives (14). This is the exact Barnes-moment evaluation the regulator r also justifies the contour endpoint before dominated passage to $r = 1$. $\square$

Lemma 13 (Initial coboundary). *One has*

$$U(0) = \begin{pmatrix} 1 & \frac{152}{5} & \frac{195477}{175} \\ 0 & \frac{1723}{90} & \frac{1963751}{2800} \\ 0 & -\frac{143}{18} & -\frac{165201}{560} \end{pmatrix}$$

and

$$AU(0)^{-T} = \frac{1}{382493} \begin{pmatrix} 30102216645 & -19896711516 & 525137760 \\ 32863650750 & -21712357800 & 573048000 \end{pmatrix}. \quad (15)$$

Proof. Substitute $n = 0$ in (10); exact inversion and multiplication by A give (15). $\square$

Lemma 14 (Exact Catalan cancellation). *The logarithmic and rational terms cancel exactly:*

$$AU(0)^{-T}\mathbf{F} = 450\sqrt{\pi} \begin{pmatrix} G \\ 1 \end{pmatrix}. \quad (16)$$

Proof. Multiply the rational matrix (15) by (14). The two coefficients of $\log 2$ are zero, the two rational constant terms are zero, and the remaining coefficients are respectively $450G$ and 450 . This is an identity in the three-dimensional $\mathbb{Q}$ -space spanned by $1, \log 2, G$; no decimal recognition is involved. $\square$

5 Exact Delannoy transform

This section repairs the finite-prefix claims in the earlier Delannoy argument. It is included because the transform and its error integral are useful exact identities. The main convergence proof below does not require a sign assertion about the transformed coefficient $f(k)$.

Let

$$\mathcal{B}(N, k) = 2^k \binom{2k}{k} \binom{N}{k} \binom{N+k}{k} \quad (0 \leq k \leq N).$$

Lemma 15 (Inverse Delannoy matrix). *The inverse of the lower-triangular matrix $\mathcal{B}$ is*

$$\beta(k, N) = \frac{(-1)^{k+N} (2N+1) \binom{k}{N}}{2^k \binom{2k}{k} (k+N+1) \binom{k+N}{N}} \quad (0 \leq N \leq k). \quad (17)$$

Proof. We prove $\sum_{N=j}^k \beta(k, N) \mathcal{B}(N, j) = \delta_{kj}$. For $k = j$ the sole product is 1. Suppose $m = k - j > 0$ and put $N = j + r$. After removing a nonzero factor independent of r and using

$$\frac{2j+2r+1}{k+j+r+1} = 2 - \frac{2k+1}{k+j+r+1},$$

the sum becomes

$$2 {}_2F_1(-m, 2j+1; k+j+1; 1) - \frac{2k+1}{k+j+1} {}_2F_1(-m, 2j+1; k+j+2; 1).$$

Chu–Vandermonde evaluates the two hypergeometric terms as

$$\frac{(m)_m}{(k+j+1)_m}, \quad \frac{(m+1)_m}{(k+j+2)_m}.$$

Their ratio is $2(k+j+1)/(2k+1)$, so the displayed difference is zero. This proves the inverse formula for every index, rather than checking a finite square. $\square$

Define the scalar normalization

$$H_N = (-16)^N (2)_N^2 (3)_N^2 \left(\frac{5}{2}\right)_N \left(\frac{7}{2}\right)_N^2$$

and, for the first column only in this section,

$$\hat{P}_N = P_{N,1}/H_N, \quad \hat{Q}_N = Q_{N,1}/H_N.$$

There are unique rational sequences f, g such that

$$\hat{Q}_N = \sum_{k=0}^N f(k) \mathcal{B}(N, k), \quad \hat{P}_N = \sum_{k=0}^N g(k) \mathcal{B}(N, k), \quad (18)$$

namely

$$f(k) = \sum_{N=0}^k \beta(k, N) \hat{Q}_N, \quad g(k) = \sum_{N=0}^k \beta(k, N) \hat{P}_N. \quad (19)$$

Theorem 16 (Holonomic k -recurrences). *The sequences $f(k)$ and $g(k)$ in (19) are P -recursive. This conclusion holds for all k as a consequence of holonomic closure applied to the exact finite sums; it is not based on a guessed operator or verification of a finite prefix.*

Proof. By (9), $\hat{P}_N$ and $\hat{Q}_N$ are coordinates of a fixed row multiplied by the three-dimensional rational first-order system $\Phi_T(N)^{-T}$. Four consecutive coordinate functionals are linearly dependent over $\mathbb{Q}(N)$; clearing denominators gives a polynomial-coefficient scalar recurrence. Thus both sequences are holonomic in N .

The kernel $\beta(k, N)$ is proper hypergeometric: both of its shift quotients are rational functions of (k, N) . Zeilberger's creative telescoping theorem [3], equivalently closure of holonomic functions under definite summation, applied to the finite range $0 \leq N \leq k$ therefore produces polynomial-coefficient recurrences for the two sums (19). The argument proves existence and annihilation identically; no order, degree, or minimality inferred by guessing is asserted. $\square$

Theorem 17 (Effective positivity and recessiveness). *For every $n \geq 0$, $F_n > 0$, and*

$$\lim_{n \rightarrow \infty} F_n^{1/n} = 17 - 12\sqrt{2} = \mu. \quad (20)$$

Consequently (F_n) is the unique recessive solution of the companion recurrence in Lemma 9.

Proof. Two beta integrals, two Laplace integrals, and the duplication formula transform the Barnes integral exactly into

$$F_n = \frac{\sqrt{\pi}}{\Gamma(n+1)\Gamma(n+2)} \int_{(0,1)^2 \times (0,\infty)^2} x^{1/2}(1-x)^n(1-y)^{n+1}u^{n+1}v^{n+1} \times \exp\left(-u-v-2\sqrt{\frac{uv}{xy}}\right) du dv dx dy. \quad (21)$$

Every factor is positive, proving $F_n > 0$ directly for every n .

For the exact root rate put $u = na$, $v = nb$ and write

$$F_n = \frac{\sqrt{\pi} n^{2n+4}}{\Gamma(n+1)\Gamma(n+2)} J_n, \quad J_n = \int_D A e^{n\Phi} dx dy da db,$$

where $D = (0,1)^2 \times (0,\infty)^2$,

$$A = x^{1/2}(1-y)ab$$

and

$$\Phi = \log(1-x) + \log(1-y) + \log a + \log b - a - b - 2\sqrt{ab/(xy)}.$$

Let $p = \sqrt{xy}$ and $r = \sqrt{ab}$. Arithmetic-geometric mean gives

$$\Phi \leq 2\log(1-p) + 2\log r - 2r(1+p^{-1}),$$

with equality only when $x = y$ and $a = b$. Maximizing first over r and then over p yields

$$\Phi + 2 \leq 2\log \frac{p(1-p)}{1+p} \leq 4\log(\sqrt{2}-1) =: \Phi_* + 2.$$

The unique maximizer is

$$x = y = \sqrt{2} - 1, \quad a = b = \frac{\sqrt{2} - 1}{\sqrt{2}}.$$

Moreover

$$Ae^\Phi \leq x^{1/2}(1-x)(1-y)^2 a^2 b^2 e^{-a-b},$$

and the right-hand side is integrable on D . Hence $J_n \leq e^{(n-1)\Phi_*} \int_D Ae^\Phi$. Conversely, on a fixed small neighborhood of the unique maximizer, A is bounded below by a positive constant and $\Phi \geq \Phi_* - \varepsilon$. It follows that $J_n^{1/n} \rightarrow e^{\Phi_*}$; this is the elementary Laplace principle with explicit maximizer, not a finite numerical check. Stirling's formula gives

$$\left(\frac{\sqrt{\pi} n^{2n+4}}{\Gamma(n+1)\Gamma(n+2)} \right)^{1/n} \rightarrow e^2.$$

Therefore

$$\lim F_n^{1/n} = e^{\Phi_*+2} = (\sqrt{2}-1)^4 = 17 - 12\sqrt{2}.$$

Lemma 10 and Proposition 4 now identify (F_n) as the unique recessive solution. $\square$

Theorem 18 (Exact integral for the Delannoy coefficient error). *Put $h(k) = g(k) - Gf(k)$ and*

$$\Psi_k(t) = \sum_{N=0}^k \beta(k, N) \frac{R_{N,1}(t^2)}{H_N}. \quad (22)$$

Then, for every $k \geq 0$,

$$h(k) = - \int_0^1 \frac{-\log t}{1+t^2} \Psi_k(t) dt. \quad (23)$$

The sum in (22) is finite and has rational coefficients; in fact

$$\Psi_k(t) = f(k) - (1+t^2)g(k). \quad (24)$$

Proof. Divide (3) by H_N , multiply by $\beta(k, N)$, and sum over $0 \leq N \leq k$. The sum has only $k+1$ terms, so its interchange with the absolutely convergent integral is automatic. Equations (19) give (23). Applying the same inverse sum to $R_{N,1}(t^2)/H_N = \widehat{Q}_N - (1+t^2)\widehat{P}_N$ gives (24). $\square$

Remark 1. Equation (24) shows why the integral identity alone must not be advertised as a decay estimate: without an independent bound it is an exact rewriting of $h(k)$. The earlier inference from finite cancellation to an all- k $O(8^{-k})$ bound is therefore deleted. The positive integral (21) supplies the effective all-index argument actually needed for the proof.

Remark 2 (Other audit corrections). If the unused raw scalar recurrence is indexed as

$$\sum_{j=0}^3 c_j(n) y_{n+j} = 0,$$

the degree pattern is

$$(28, 21, 14, 7), \quad \deg c_j = 7(4-j),$$

not $\deg c_j = 7(3-j)$. The former modular paragraph has also been removed: the eta product $\eta^2(2\tau)\eta^2(4\tau)$ has formal weight 2, and no modular claim is required here. All endpoint passages used above are made by dominated convergence or by the positive Laplace integral.

6 Limit identification and all three columns

Theorem 19 (Hatted sequences and the three-column bridge). *For every $N \geq 0$ and $j = 1, 2, 3$, define*

$$\hat{P}_{N,j} = \frac{P_{N,j}}{H_N}, \quad \hat{Q}_{N,j} = \frac{Q_{N,j}}{H_N}, \quad H_N = (-16)^N (2)_N^2 (3)_N^2 \left(\frac{5}{2}\right)_N \left(\frac{7}{2}\right)_N^2. \quad (25)$$

Then

$$\frac{P_{N,j}}{Q_{N,j}} = \frac{\hat{P}_{N,j}}{\hat{Q}_{N,j}} \longrightarrow G \quad (j = 1, 2, 3).$$

Proof. By (9), for each standard column e_j ,

$$\begin{pmatrix} \hat{P}_{N,j} \\ \hat{Q}_{N,j} \end{pmatrix} = A\Phi_T(N)^{-T} e_j. \quad (26)$$

Theorem 17 identifies $\mathbf{F}$ as the recessive initial vector. Lemma 11 verifies the nonzero terminal coefficient separately for each of $j = 1, 2, 3$. Proposition 4 and Lemma 14 therefore give

$$[A\Phi_T(N)^{-T} e_j] \longrightarrow [AU(0)^{-T} \mathbf{F}] = \left[450\sqrt{\pi} \begin{pmatrix} G \\ 1 \end{pmatrix} \right] = \left[\begin{pmatrix} G \\ 1 \end{pmatrix} \right].$$

The second coordinate of a normalized representative tends to the nonzero number 1, so $\hat{Q}_{N,j}$ (and hence $Q_{N,j}$) is nonzero for all sufficiently large N , and $\hat{P}_{N,j}/\hat{Q}_{N,j} \rightarrow G$. Finally $H_N \neq 0$, so the equality of the hatted and original ratios in (25) proves all three assertions of Theorem 1. $\square$

7 An independent proof by an elementary Miller cone

This section gives a second proof of the main assertion. It uses only the printed matrix, elementary real integrals, rational identities, and a uniform contraction argument. In particular, it does not use a Meijer- G function, a Barnes integral, or a special-function connection formula.

7.1 The common projective limit

Put $S = \text{diag}(1, -1, 1)$ and

$$B_n = -SM(n)S.$$

Every entry of B_n is positive for $n \geq 0$. If $\mathbf{p}_n, \mathbf{q}_n$ are the original numerator and denominator rows, define

$$\tilde{\mathbf{p}}_n = (-1)^n \mathbf{p}_n S, \quad \tilde{\mathbf{q}}_n = (-1)^n \mathbf{q}_n S.$$

Then

$$\tilde{\mathbf{p}}_{n+1} = \tilde{\mathbf{p}}_n B_n, \quad \tilde{\mathbf{q}}_{n+1} = \tilde{\mathbf{q}}_n B_n,$$

and the signs cancel in the quotient:

$$\frac{\tilde{p}_{n,j}}{\tilde{q}_{n,j}} = \frac{P_{n,j}}{Q_{n,j}}.$$

The following elementary contraction will also be used at the end.

Lemma 20 (Common-limit lemma). *There is a real number L such that*

$$\frac{P_{n,j}}{Q_{n,j}} \rightarrow L \quad (j = 1, 2, 3).$$

More precisely, all three ratios lie in an interval whose length is $O((2/3)^n)$.

Proof. Set

$$X_n = (n+1) \frac{\tilde{q}_{n,2}}{\tilde{q}_{n,1}}, \quad Y_n = (n+1)^3 \frac{\tilde{q}_{n,3}}{\tilde{q}_{n,1}}.$$

Direct substitution of the entries of B_n gives the invariant rectangle

$$1 \leq X_n \leq \frac{3}{2}, \quad 0 \leq Y_n \leq 2. \quad (27)$$

Here is a short exact way to check the induction. After the common positive factor $\tilde{q}_{n,1}$ is removed, each new coordinate is affine in (X_n, Y_n) . Each of the four required inequalities is therefore checked at the four corners

$$(1, 0), (1, 2), (3/2, 0), (3/2, 2).$$

After clearing the positive denominators, the resulting expressions are polynomials in n with non-negative coefficients.

Let $r_{n,i} = \tilde{p}_{n,i}/\tilde{q}_{n,i}$. Positivity of B_n writes each $r_{n+1,j}$ as a convex combination of the three $r_{n,i}$. Using (27), the normalized weights of both $r_{n,1}$ and $r_{n,2}$ are at least $1/6$. Thus, if $\ell_n \leq r_{n,i} \leq u_n$ for all i , then

$$\frac{r_{n,1} + r_{n,2}}{6} + \frac{2}{3}\ell_n \leq r_{n+1,j} \leq \frac{r_{n,1} + r_{n,2}}{6} + \frac{2}{3}u_n.$$

Defining the next enclosing interval by these two endpoints gives $u_{n+1} - \ell_{n+1} = (2/3)(u_n - \ell_n)$. The nested lower and upper endpoints consequently have one common limit, and they squeeze every $r_{n,j}$. $\square$

7.2 A positive adjoint moment

For $p, q, v \in (0, 1)$ put

$$D = pq(1 + v^2) + 2v$$

and define

$$I_n(A, B, C; k) = \iiint_{(0,1)^3} \frac{16p^{2n+6+A}q^{2n+5+B}(1-p^2)^n(1-q^2)^{n+1}v^{2n+3+C}}{D^{2n+4+k}} dp dq dv. \quad (28)$$

All these integrals are finite and positive. Define

$$\begin{aligned} w_{n,1} &= I_n(0, 0, 0; 0), \\ w_{n,2} &= 2(n+2)I_n(0, 0, 1; 1), \\ w_{n,3} &= -(n+2)I_n(0, 0, 1; 1) + 2(n+2)(2n+5)I_n(0, 0, 2; 2). \end{aligned} \quad (29)$$

The last component is also positive. One proof is to integrate in v after applying integration by parts to a multiple of $v^{2n+4}(1-v^2)/D^{2n+5}$; the boundary terms vanish and the remaining integrand is positive.

Lemma 21 (Adjoint recurrence). *For*

$$\lambda_n = (n+1)(n+2)^2(n+3)^2(2n+7)^2$$

the vector in (29) satisfies

$$B_n w_{n+1} = \lambda_n w_n. \quad (30)$$

Moreover

$$0 < w_{n,1} \leq \frac{512}{64^n}. \quad (31)$$

Proof. The recurrence has a three-variable divergence certificate. To state the certificate compactly, for a polynomial $H = H(n, p, q, v)$ define

$$\begin{aligned} \mathcal{L}_p(H) &= D\{p(1-p^2)H_p + [(2n+7) - (4n+9)p^2]H\} \\ &\quad - (2n+7)p(1-p^2)q(1+v^2)H, \\ \mathcal{L}_q(H) &= D\{q(1-q^2)H_q + [(2n+6) - (4n+10)q^2]H\} \\ &\quad - (2n+7)q(1-q^2)p(1+v^2)H, \\ \mathcal{L}_v(H) &= D\{v(1-v^2)H_v + [(2n+4) - (2n+6)v^2]H\} \\ &\quad - (2n+7)v(1-v^2)(2pqv+2)H. \end{aligned}$$

For each of the three rows there are explicit polynomials $H_{i,p}, H_{i,q}, H_{i,v} \in \mathbb{Z}[n, p, q, v]$ such that, after the base integrand in (28) is factored out,

$$4(2n+3)(n+2) \left(\sum_{j=1}^3 (B_n)_{ij} W_{n+1,j} - \lambda_n W_{n,i} \right) = \mathcal{L}_p(H_{i,p}) + \mathcal{L}_q(H_{i,q}) + \mathcal{L}_v(H_{i,v}). \quad (32)$$

Here $W_{n,i}$ denotes the rational factor which produces the i th integrand in (29). The sparse integer coefficient lists for the nine polynomials are included in the accompanying Lean source. Its kernel checks (32) as a polynomial identity, not by sampling. Integration of the right side is zero: the three antiderivatives contain, respectively, $p(1-p^2)$, $q(1-q^2)$, and $v(1-v^2)$, so both endpoint terms vanish. This proves (30).

For (31), rewrite the first integrand as

$$\frac{512}{64^n} p(1-q^2)[4p^2(1-p^2)]^n [4q^2(1-q^2)]^n \left(\frac{pq}{D}\right)^5 \left(\frac{2v}{D}\right)^{2n+3} \left(\frac{D}{4}\right)^4.$$

Every factor after $512/64^n$ belongs to $[0, 1]$ on the unit cube. The cube has volume one, giving the required bound. $\square$

At the initial index the same moments select Catalan's constant exactly. Writing

$$\tilde{\mathbf{p}}_0 = (30921, 32972, 8240), \quad \tilde{\mathbf{q}}_0 = (33750, 36000, 9000),$$

one has

$$\tilde{\mathbf{q}}_0 \cdot \mathbf{w}_0 = 75, \quad \tilde{\mathbf{p}}_0 \cdot \mathbf{w}_0 = 75G. \quad (33)$$

These are elementary integral evaluations. Expanding (29), the two coefficient triples are

$$(33750, 126000, 180000), \quad (30921, 115408, 164800).$$

Fubini reduction first gives $75/16$ and $75G/16$ before the outer factor 16 in (28). The remaining one-variable substitution $t = 2v/(1 + v^2)$ reduces the second evaluation to

$$G = \int_0^1 \frac{-\log t}{1 + t^2} dt.$$

All endpoint terms are rational functions and logarithms; hence (33) is an exact evaluation, not numerical recognition.

7.3 A rational companion and its Miller graph

Let

$$\sigma_n = -2(n+2)^2(n+3)^2(2n+5)(2n+7)^2, \quad T_n = \sigma_n M(n)^{-T}.$$

Thus $T_n^T M(n) = \sigma_n I$. With $e_1 = (1, 0, 0)^T$, form the Krylov matrix

$$K_n = [e_1, T_n e_1, T_n T_{n+1} e_1].$$

Direct rational simplification gives

$$T_n K_{n+1} = K_n C_n, \quad C_n = \begin{pmatrix} 0 & 0 & c_{0,n} \\ 1 & 0 & c_{1,n} \\ 0 & 1 & c_{2,n} \end{pmatrix}. \quad (34)$$

For completeness, the coefficients are

$$\begin{aligned} c_{0,n} &= \frac{(n+2)^2(2n+7)d(n+1)}{(n+4)(n+5)(2n+9)d(n)}, \\ c_{1,n} &= -\frac{\pi(n)}{(n+4)(n+5)(2n+5)(2n+9)d(n)}, \\ c_{2,n} &= \frac{\omega(n)}{4(n+3)(n+5)(2n+7)(2n+9)d(n)}, \end{aligned} \quad (35)$$

where

$$\begin{aligned} d(n) &= 3072n^7 + 61824n^6 + 531184n^5 + 2525522n^4 \\ &\quad + 7175793n^3 + 12183785n^2 + 11446123n + 4589916, \\ \pi(n) &= 430080n^{11} + 15954432n^{10} + 267573056n^9 \\ &\quad + 2677696216n^8 \\ &\quad + 17764567836n^7 + 82030514302n^6 + 269008975593n^5 \\ &\quad + 626475469357n^4 + 1015303508704n^3 + 1090544343906n^2 \\ &\quad + 698695833276n + 202285840752, \\ \omega(n) &= 1720320n^{11} + 62195712n^{10} + 1017208064n^9 \\ &\quad + 9933863520n^8 \\ &\quad + 64362354608n^7 + 290489793800n^6 + 931925873076n^5 \\ &\quad + 2125092069586n^4 + 3375535586927n^3 + 3557018693358n^2 \\ &\quad + 2237964273360n + 636907494879. \end{aligned}$$

Clearing the positive denominators in (35) gives polynomials with nonnegative coefficients and proves, for every $n \geq 0$,

$$1 \leq c_{0,n} \leq \frac{4}{3}, \quad -49 \leq c_{1,n} \leq -35, \quad 35 \leq c_{2,n} \leq 37. \quad (36)$$

Set

$$A_n = -\frac{c_{1,n}}{c_{0,n}}, \quad B'_n = -\frac{c_{2,n}}{c_{0,n}}, \quad R_n = \frac{1}{c_{0,n}}.$$

Then

$$\frac{105}{4} \leq A_n \leq 49, \quad -37 \leq B'_n \leq -\frac{105}{4}, \quad \frac{3}{4} \leq R_n \leq 1. \quad (37)$$

Let

$$\mathcal{K} = [0, 1/16] \times [0, 1/256]$$

and define the backward projective map

$$\mathcal{P}_n(u, v) = \left(\frac{1}{A_n + B'_n u + R_n v}, \frac{u}{A_n + B'_n u + R_n v} \right). \quad (38)$$

Equations (37) imply

$$23 \leq A_n + B'_n u + R_n v \leq 50 \quad ((u, v) \in \mathcal{K}).$$

Consequently $\mathcal{P}_n(\mathcal{K}) \subseteq \mathcal{K}$, and a direct difference estimate gives

$$\|\mathcal{P}_n(x) - \mathcal{P}_n(y)\|_\infty \leq \frac{1}{8} \|x - y\|_\infty. \quad (39)$$

Thus there is a unique infinite pullback graph $z_n = \mathcal{P}_n(z_{n+1})$ with $z_n \in \mathcal{K}$; truncating at depth h changes z_n by at most 8^{-h} times the diameter of $\mathcal{K}$.

It remains to show that the positive moment is precisely the solution on this graph. Put $r_n = Sw_n$ and

$$s_n = -\frac{\sigma_n}{\lambda_n} = \frac{2(2n+5)}{n+1} \leq 10, \quad a_0 = 1, \quad a_{n+1} = a_n s_n, \quad f_n = a_n w_{n,1}.$$

The adjoint identity and (34) imply that

$$F_n = (f_n, f_{n+1}, f_{n+2})$$

obeys the companion evolution $F_{n+1} = F_n C_n$. Moreover

$$|f_n| \leq 512(5/32)^n. \quad (40)$$

For any companion solution $X = (X_0, X_1, X_2)$ define its defect from the graph by

$$\delta_n(X) = (X_1 - z_{n,1}X_0, X_2 - z_{n,2}X_0).$$

Using $z_n = \mathcal{P}_n(z_{n+1})$ and (34),

$$\begin{aligned} \delta_{n+1,1} &= -z_{n+1,1}\delta_{n,1} + \delta_{n,2}, \\ \delta_{n+1,2} &= (c_{1,n} - z_{n+1,2})\delta_{n,1} + c_{2,n}\delta_{n,2}. \end{aligned}$$

The bounds above give

$$\|\delta_{n+1}(X)\|_\infty \geq \frac{1}{2} \|\delta_n(X)\|_\infty. \quad (41)$$

On the other hand (40) gives

$$2^n \|\delta_n(F_n)\|_\infty \leq 544(5/16)^n \longrightarrow 0.$$

Iterating (41) backwards forces $\delta_0(F_0) = 0$:

$$F_0 = f_0(1, z_{0,1}, z_{0,2}). \quad (42)$$

This is the elementary Miller selection theorem needed here.

7.4 Initial selection and the literal terminal column

The initial Krylov matrix and two fixed covectors are

$$K_0 = \begin{pmatrix} 1 & 152/5 & 195477/175 \\ 0 & 1723/90 & 1963751/2800 \\ 0 & -143/18 & -165201/560 \end{pmatrix},$$

$$\ell_P = \begin{pmatrix} 30102216645 \\ -19896711516 \\ 525137760 \end{pmatrix}, \quad \ell_Q = \begin{pmatrix} 32863650750 \\ -21712357800 \\ 573048000 \end{pmatrix}.$$

Exact multiplication gives

$$K_0 \ell_P = 382493 \mathbf{p}_0^T, \quad K_0 \ell_Q = 382493 \mathbf{q}_0^T. \quad (43)$$

Together with (33) and (42), this yields

$$\frac{\ell_P \cdot (1, z_{0,1}, z_{0,2})}{\ell_Q \cdot (1, z_{0,1}, z_{0,2})} = G. \quad (44)$$

The denominator is positive throughout $\mathcal{K}$; this is immediate by checking the two relevant endpoints of its affine expression.

We finally connect the abstract pullback to an actual column of the challenge product. Put $A_n = C_n^{-T}$. The identities already proved give the exact telescope

$$K_0^T \mathcal{M}_N = \Sigma_N (A_0 A_1 \cdots A_{N-1}) K_N^T \quad (45)$$

for a nonzero scalar Σ_N . For $N = h + 2$, direct rational simplification gives

$$A_h A_{h+1} K_N^T e_3 = \kappa_N (1, \tau_N, 0)^T, \quad \kappa_N > 0, \quad (46)$$

where

$$\begin{aligned} V_N &= N^2(2N+1)(2N+3)(48N^4 + 306N^3 + 715N^2 + 729N + 275), \\ R_N &= 6528N^8 + 54480N^7 + 187144N^6 + 344616N^5 \\ &\quad + 371338N^4 + 239883N^3 + 91396N^2 + 19200N + 1800, \quad \tau_N = \frac{V_N}{R_N}. \end{aligned}$$

Clearly $\tau_N \geq 0$, while $\tau_N \leq 1/16$ follows after clearing the positive denominator from

$$\begin{aligned} R_N - 16V_N &= 3456N^8 + 28752N^7 + 99912N^6 + 191752N^5 + 226106N^4 \\ &\quad + 169691N^3 + 78196N^2 + 19200N + 1800 \geq 0, \end{aligned}$$

where R_N is the denominator displayed in τ_N . Hence $(\tau_N, 0) \in \mathcal{K}$.

Applying (45) to e_3 and cancelling its nonzero scalar factors gives

$$\frac{P_{h+2,3}}{Q_{h+2,3}} = \frac{\ell_P \cdot (1, \mathcal{P}_0 \mathcal{P}_1 \cdots \mathcal{P}_{h-1}(\tau_{h+2}, 0))}{\ell_Q \cdot (1, \mathcal{P}_0 \mathcal{P}_1 \cdots \mathcal{P}_{h-1}(\tau_{h+2}, 0))}.$$

By (39), its chart tends to z_0 ; by (44), the quotient tends to G . Lemma 20 then promotes the third-column limit to all three columns. This proves the main assertion independently.

Formal verification. The accompanying Lean development proves every analytic estimate, polynomial certificate, matrix identity, and limit used in this section. Its final theorem is

`problem25_solved : Problem25Claim.`

It contains no `sorry`. Its axiom audit is

`{propext, Classical.choice, Quot.sound}`.

Remark 3 (Provenance of the two proofs). The Meijer- G route above was reconstructed after publication of the official solutions and uses their special-function trajectory. The proof in Section 7 was developed independently from the printed challenge matrix. It uses no Meijer- G or Barnes connection data.

References

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